Case study: assessing grammaticality judgement data on matrix questions in Brazilian Portuguese

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1 Introduction

Null Subject Languages (henceforth, NSL) or pro-drop[1] languages have been a focus of investigation in the last four decades (Rizzi 1986; Jaeggli and Safir 1989; Alexiadou and Anagnostopoulou 1998; Barbosa et al. 2005; Barbosa 2009; Biberauer et al. 2009 among many others). Within the general inquiry on the properties of NSLs, a subgroup of languages, so called partial NSLs, has attracted considerable attention (Holmberg, 2005; Holmberg et al., 2009; Holmberg, 2010; Barbosa, 2019). Partial NSLs are generally understood as pro-drop languages where subject drop is restricted to specific syntactic environments (e.g., subject drop is only permitted in embedded clauses). In that respect, probing into the structural conditions behind pro-drop in partial NSLs can shed light on the nature of empty categories and their corresponding structural conditions within and outside narrow syntax.

In this study I review a subset of data from one partial NSL, Brazilian Portuguese (henceforth, BP), which poses problems for current approaches of partial pro-drop. Namely, I focus on

[1]Throughout this article the terms "null subject" and "pro-drop" will be used interchangeably. While pro-drop goes beyond null subjects (e.g., object drop) the present study is focused entirely on subject drop.

pro-drop in matrix question clauses in BP and the attested interaction between pro-drop and wh-movement in this syntactic environment (Modesto, 2000; Ferreira, 2009; Rodrigues, 2004; Nunes, 2008). This subset of data poses problems for current approaches to parial NSLs and to specific analyses centered around BP. In that context, the patterns I am interested in this article have important theoretical implications, as they seem to require some degree of reformulation of current theories.

My focus on this article is assessing whether the patterns attested for matrix pro-drop in BP are empirically robust. Besides the potential theoretical implications just outlined, the motivation for reassessing this subset of data is twofold. One, it has been extensively observed that BP displays a high degree of interspeaker variability (Lucchesi 2006; Galves 1985; Zilles 2005; Poplack and Malvar 2007; Avelar 2006, among many others). Secondly, it is well known that pro-drop in BP is part of a not fully understood ongoing change in the language (Kato, 2000; Kato and Negrão, 2000; Nunes, 2011; Duarte, 1993, 2000).

Given the aforementioned facts, experimentally assessing the reported judgements for matrix pro-drop in BP may yield valuable information. In recent literature (Sprouse et al., 2016; Almeida, 2014) it has been shown that the usage of formal experimental methods for collecting grammaticality judgement data can expand the degree of understanding of syntactic data.

For that purpose I have conducted two grammaticality judgement experiments with two large (>200 participants) sample of speakers. The results of the study point to matrix clauses in BP showing no significant grammaticality effect associeted to i) pro-drop, ii) wh-movement, iii) the interaction between pro-drop and wh-movement and iv) number features. However, the experimental results point to a general dispreferrence, within grammatical structures, for pro-drop and for pro-drop in the context of wh-movement.

This article is organized as follows. Section 2 provides an overview of the current literature on partial NSLs while also providing the relevant data and the general theoretical problem. Section 3 describes the methodology used in this study. Section 4 presents the experimental results and

statistical analysis. Section 5 is a discussion of the results in the context of previous literature. Section 6 offers concluding remarks and prospects for future work.

2 Outline

The main empirical object of this study is a subset of data on pro-drop in matrix questions in BP. The reason this data is relevant is twofold. One, it poses an empirical problem for generalized theories[2] of partial pro-drop. Secondly, the patterns I am concerned with are one of the main arguments of a second group of analyses focused specifically on BP, which I refer to as "BP specific analyses". In that regard, the main goal of this study is assessing if these patterns are as empirically sound as reported and if the BP specific analyses may be extended to the generalized accounts of partial pro-drop.

For the generalized theories of partial pro drop (Holmberg, 2005, 2010; Barbosa, 2019) matrix pro-drop - in BP and in other partial NSLs, such as Finnish and Hebrew - is considered marginal or exceptional phenomena. These theories are largely concerned with the more commonly attested and better understood instances of partial pro-drop, that is, pro-drop in embedded clauses, where pro-drop behaves much like a bound trace 'controlled' by a matrix antecedent. Example (1) illustrates embedded pro-drop in BP. In (1) we see pro-drop for 1st, 2nd and 3rd person subjects with the caveat that all instances of pro-drop require a matrix co-referent subject antecedent, as shown in (1)(c) where two potential antecedents exist - Carlos and Pedro - but only the subject, Pedro, can 'control' the embedded null subject. The pattern in (1) is crosslinguistically well attested: other partial NSLs (e.g., Finnish, Hebrew) display similar patterns.

[2]By "generalized theories" I refer to theories that are not language-specific but attempt to cross-linguistic account of partial pro-drop). These theories contrast with specific theories focused on language internal patterns, like the BP specific theories I refer to in later paragraphs.

(1) **Embedded null subjects in Brazilian Portuguese**

a. O Giancarlo$_i$ acha que (ele$_{i/*k}$) vai comprar uma casa.
DET.M Giancarlo think.3SG that (he) go.3SG.FUT buy one house
"Giancarlo thinks that Giancarlo will buy a house."

b. Eu acho que (eu) vou comprar uma casa.
I think.1SG that (I) go.1SG.FUT buy one house
"I think that I will buy a house."

c. Carlos$_i$ disse para o Pedro$_j$ que (ele$_i$) ficou rico.
Carlos$_i$ say.3SG.PST to DET.M Peter$_j$ that (he$_{i/*j/*k}$) get.3SG.PST rich.SG.M
"Carlos told Peter that Carlos got rich."

The generalized theories of partial pro-drop offer different approaches to the pattern illustrated above. While they might diverge in how their proposals play out, all theories of partial pro-drop rely on some structural relation between an underlying non-overt syntactic category which is anchored, in some form, to a higher and overt nominal. This line of explanation fares well for embedded pro-drop but does not extend clearly to matrix pro-drop, as I will show. Before moving on it is important to notice that the generalized theories of partial pro-drop do acknowledge some instances of matrix pro-drop, but they consider these marginal or exceptional phenomena which are not part of their core theoretical explanation. Barbosa (2019), for example, considers matrix pro-drop in BP an instance of exempted anaphora. Similalry, Holmberg (2005) proposes that matrix pro-drop in partial NSLs are fully feature-specified covert nominals, unlike embedded pro-drop which is accounted for by a control-like relation between a bundle of unspecified features and an overt nominal.

Let us consider now the reported grammaticality judgement BP data on matrix pro-drop to assess where the potential empirical gaps lie within generalized theories. Only 1st person subjects may be dropped in declarative clauses in BP, as shown in examples in (2) (Modesto, 2000; Ferreira, 2009; Rodrigues, 2004; Nunes, 2008). This pattern is not unique: similar patterns are attested in Hebrew and Finnish.

(2) **Only 1st person subjects can be dropped in matrix clauses in BP**

a. **(Eu)** cheguei.
(I) arrived.1SG
"I arrived."

b. **(Nós)** chegamos.
(We) arrived.1PL
"We arrived."

c. *(Você) chegou.
(you) arrived.2SG
"You arrived."

d. *(Vocês) chegaram.
(you-P) arrived.2PL
"You arrived."

e. *(Ela/ele) chegou.
(she/he) arrived.3SG
"She/he arrived."

f. *(Elas/eles) chegaram.
they-F/they-M arrived.3PL
"They arrived."

While the pattern in (2) is acknowledged by generalized partial pro-drop analyses, the patterns becomes more complex in question environments. 1st *and* 2nd person subjects may be dropped in yes/no questions, as shown in (3).

(3) **1st and 2nd person subjects can be dropped in yes/no questions**

a. (Eu) dormi muito ontem?
I slept.1SG much yesterday?
"Have I slept for a long time last night?"

b. (Você) dormiu muito ontem?
You slept.2SG much yesterday?
"Have you slept for a long time last night?"

c. *(Ele/ela) dormiu muito ontem?
He/she slept.3SG much yesterday?
”Has she/he slept for a long time last night?”

The pattern seen in yes/no questions extends to wh-questions with wh-in-situ, as seen in (4).

(4) **1st and 2nd person subjects can be dropped in *wh*-questions with *wh*-in-situ**

a. (Eu) tô fazendo o que?
(I) be.1SG doing DET.M what
”What am I doing?”

b. (Nós) estamos fazendo o que?
(We) be.1PL doing DET.M what
”What are we doing?”

c. (Você) tá fazendo o que?
(you) be.2SG doing DET.M what
”What are you doing?”

d. (Vocês) tão fazendo o que?
(you-P) be.2PL doing DET.M what
”What are you doing?”

e. *(Ela/Ele) tá fazendo o que?
(she/he) be.3SG doing DET.M what
”What is she/he doing?”

f. *(Elas/Eles) tão fazendo o que?
(they-F/they-M) be.3PL doing DET.M what
”What are they doing?”

The data in (4) presents wh-questions with wh-in-situ. BP is not a strict wh-in-situ language: wh-movement is also possible, and there is an ongoing discussion on whether or not wh-movement is truly optional in the language (Yeo, 2011; Kato, 2014; SILVA and Grolla, 2016). While the distribution of null subjects in questions with wh-in-situ matches the distribution of yes/no questions,

wh-movement introduces a strikingly different pattern. As seen in (5), when wh-movement is overt[3], pro-drop is always reported as ungrammatical.

(5) **Subject drop is ungrammatical in *wh*-questions with *wh*-movement**

a. **O** **que**$_i$ *(eu) tô fazendo **t**$_i$?
DET.M what *(I) be.1SG doing?
"What am I doing?"

b. **O** **que**$_i$ *(nós) estamos fazendo **t**$_i$?
DET.M what *(I) be.1PL doing?
"What are we doing?"

c. **O** **que**$_i$ *(você) tá fazendo **t**$_i$**?**
DET.M what *(you) be.2SG doing?
"What are you doing?"

d. **O** **que**$_i$ *(vocês) estão fazendo **t**$_i$**?**
DET.M what *(you.PL) be.2PL doing?
"What are you doing?"

The contrast between (3), (4) and (5) shows that 1st and 2nd person null subjects are fine in matrix yes/no questions and wh-questions with wh-in-situ, but ungrammatical when there is wh-movement. The contrast between yes/no and wh-in-situ questions on the one hand and questions with overt wh-movement on the the other hand is entirely unexpected under current generalized theories of partial pro-drop.

Since generalized theories of partial pro-drop consider matrix pro-drop exceptional and embedded pro-drop the canonical phenomenon, their accounts ignore the core structural properties of pro-drop in matrix clauses. Specifically, these accounts overlook the apparent interaction between pro-drop and wh-movement in BP. This interaction is, however, accounted for by BP specific accounts (Modesto, 2000; Ferreira, 2009; Rodrigues, 2004; Nunes, 2008). These authors propose that matrix pro-drop in BP can be reduced to topic drop. While authors diverge when

[3]I assume here, following the literature on wh-in-situ, that wh-in-situ always involves some kind of covert movement, e.g., operator movement or feature movement. The precise mechanics of covert movement are not relevant for this study

it comes to the syntactic mechanisms behind topic drop, all agree in that matrix pro-drop in BP is an empty category bound by a higher topicalized element at the left periphery which is not overtly realized. These analyses report the judgements just presented in this study and rely on the interaction between pro-drop and wh-movement to argue for topic-drop. Acording to this group of authors, wh-moved elements create a minimality violation between the topic probe at the Left Periphery and the topicalized element. This minimality violation stops topicalization altogether, thus blocking topic drop and in consequence matrix pro-drop.

In that regard, we have a subset of data which presents challenges for the broad theories while being accounted for by language-specific proposals. The goal of this study is assessing how solid are the reported judgements supporting the language-specific proposals. The theoretical motivation for doing so is the fact that if the language-specific proposals are correct, then some reformulation of the generalized theories is necessary.

2.1 Interim Summary

To summarize the discussion thus far: a subset of data in the distribution of pro-drop in BP seems unaccounted for in current literature. Namely, pro-drop in matrix clauses and the apparent interaction between pro-drop and wh-movement present a problem for current approaches. The distribution of pro-drop in matrix clauses in summarized in Table 1.

	Declarative	Y/N Question	Wh-question (in-situ)	Wh-question (movement)
1st person	✓	✓	✓	**x**
2nd person	✓	✓	✓	**x**
3rd person	**x**	**x**	**x**	**x**

Table 1: The distribution of matrix pro-drop in BP

While the pattern just presented has been attested through informal judgement data there is a considerable degree of inter-speaker variability in judgements. Some speakers, for example,

judge that while pro-drop in the presence of wh-movement sounds ”unnatural” it is not ungrammatical, while others provide judgements of complete ungrammaticality. That we see judgement variability is unsurprising in the face of the rich literature on language variation in BP (Silva-Corvalán, 1994; Lucchesi, 2006; Scherre and Naro, 2006; Freitag et al., 2012).

Moreover, it is well known that the distribution of pro-drop in BP is the result of a not fully understood historical process (Kato, 2000; Kato and Negrão, 2000; Nunes, 2011; Duarte, 1993, 2000) where BP emerged from its original super-extract, European Portuguese, a consistent NSL language.

In this context, Duarte (1993) conducted a corpus study of plays and newspaper articles published in Brazil throughout the mid 19th century to the late 20th century. Her goal was observing how overt pronoun usage changed in this period. Duarte reports the rate of overt subjects around 20% in 1845. This number increases steadily in the following decades, reaching 74% in 1992. Duarte argues the increase in overt subjects shows an increasing dispreferrence for pro-drop in BP, one she attributes - following most of the literature - to the morphological impoverishment of BP’s verbal morphology.

Crucial for this study is the fact that the empirical landscape around matrix pro-drop in BP may be more complex than the reported judgement data suggests. For example, the distribution of pro-drop in BP may be directly related to extra-grammatical (e.g., sociolinguistic, diachronic) factors. These factors may directly affect speakers judgements, but the full extent of their potential effects is not accounted for or acknowledged in the generative literature on pro-drop. Current literature reports judgements exclusively on a grammatical versus non-grammatical paradigm which may not reflect the granularity of speakers’ judgements in face of the factors just outline.

While it has been shown that informal methods of data collection in linguistics are reliable (Sprouse et al., 2013; Sprouse and Almeida, 2013), the usage of experimental methods has proved to reveal more granular data patterns which are potentially useful for the better formulation of syntactic theory Sprouse et al. (2016). In that regard, the BP data presented in this study seems

like a good candidate for assessing the robustness of the reported patterns.

A second point of interest concerns number features and pro-drop. While pro-drop in matrix clauses has been associated with person features (e.g., the contrast between 1st/2nd person and 3rd person) there has been no studies on the role of number and its potential relation to subject drop in BP. In that context, probing into the potential interaction of number within the distribution of matrix pro-drop might reveal potentially crucial information on the nature of pro-drop in BP.

For that end I have designed two experiments to probe into speakers' judgements around matrix pro-drop and its interactions with wh-movement and number. In the next section I go over the full experimental design and the theoretical assumptions motivating it.

3 Methods

3.1 General design

The general experimental paradigm in this study mirrors Sprouse et al. (2016) factorial design for grammaticality judgement data collecting. Broadly speaking, both experiments consisted in surveys completed by native speakers of BP. Experiment 1 targeted first person subjects and Experiment 2 targeted second person subjects. Each survey contained a list of sentences which were shown one at a time to participants together with a scale from 1 do 7. To collect grammaticality data, participants were asked to rate each sentence using the scale. Participants were informed that a rating of 1 corresponded to a completely unacceptable sentence and a rating of 7 to a completely acceptable sentence. Each list was taken at random from a set of 8 randomized lists to control for potential biases. Each list contained 24 sentences, 16 of them being experimental fillers and 8 being target sentences with the relevant structures. Each of the 8 target sentences contained a combination of the 2 experimental factors. All target sentences were lexically matched.

The goal of each experiment was collecting judgement data from participants for 8 differ-

ent sentence types. Both experiments were designed in the PennController for IBEX (PCIBEX) platform (Zehr, 2018).

3.2 Factors

The factors used to design the experiments were SUBJECT OVERTNESS, WH-MOVEMENT and NUMBER. The first two factors reflect naturally the effect we want to assess: the interaction between pro-drop and wh-movement. NUMBER is included as a factor since we are also interested in checking whether number (singular/plural) plays any role in the distribution of pro-drop.

The factor SUBJECT OVERTNESS has two values: one where the subject is dropped, NULL and one where it is not, OVERT. Similarly, wh has the values WHMOV and INSITU. Finally, NUMBER has the values SINGULAR and PLURAL. The 2x2x2 design gives us 8 target sentences, henceforth referred to as conditions.

Using the 2x2x2 factorial design just outlined, two experiments were designed to collected grammatically judgement data. Experiment 1 targeted exclusively 1st person subjects and Experiment 2 targeted 2nd person subjects. The distribution was broken down into two experiments to avoid adding more than 3 factors per experiment, increasing significantly the number of experimental items per survey. The general experimental design and initial predictions - which will be further explored in the next section - are summarized in Table 2 below, where cells with LOW indicate expected low ratings from speakers (e.g., ungrammaticality, as informed by the literature) and cells with HIGH indicate expected high ratings (e.g., grammaticality).

3.3 Hypotheses

Following the literature, we expect the lowest ratings for all conditions where we have both pro-drop and wh-movement, as those are reported as ungrammatical. Every other experimental

condition is expected to be rated as grammatical, e.g., with values at the upper end of the scale. Importantly, we expect the ungrammatical condition to be lower than every other condition, e.g., a super-additivity effect in relation to other conditions. Moreover, since we expect number to play no role in pro-drop in BP, our hypothesis is that both singular and plural subjects will be rated equally by speakers. Both hypotheses are formulated below.

(6) **Hypothesis A: pro-drop and wh-movement**

We expect the lowest judgements for the NULL + WHMOV condition

(7) **Hypothesis B: number and pro-drop**

We expect number to play no role in increasing or reducing judgements for all conditions.

Concerning all other factors we will remain agnostic. Since the literature reports pro-drop for 1st and 2nd person subjects as grammatical, we will not formulate hypotheses about subject overtness *without* wh-movement. We will assume for this study that whenever pro-drop is grammatical it causes no effects in judgements. Similarly, we will assume that person plays no role in pro-drop when it comes to comparing 1st and 2nd person. Since both 1st and 2nd person pro-drop are reported as equally grammatical in wh-questions, we assume that both Experiment 1 and Experiment 2 will have provide similar results.

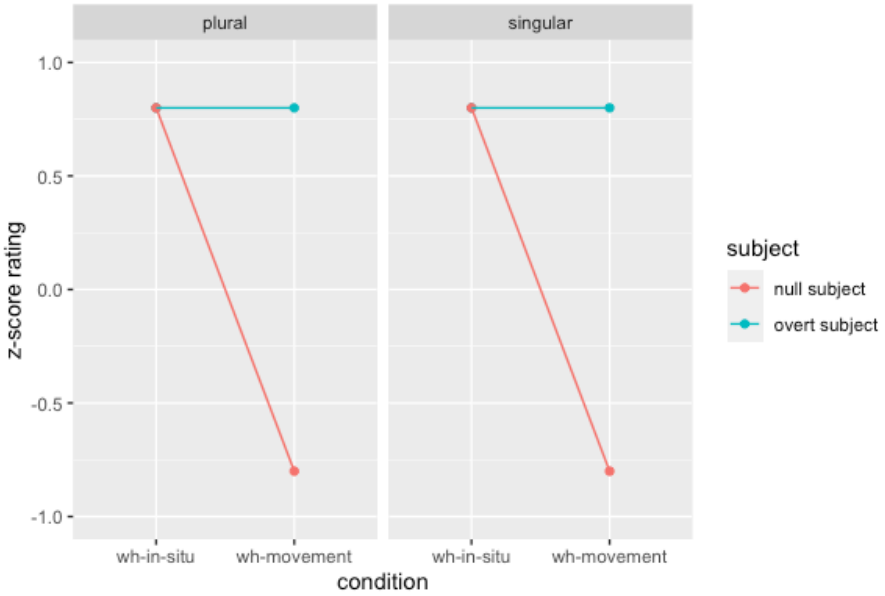
Hypothesis A and B together allows us to picture the data we expect to find. If we expect pro-drop + wh-movement to be the lowest rated condition, number and person to play no role in

Subject	Singular	Plural
Null + whmov	LOW	LOW
Null + insitu	HIGH	HIGH
Overt + whmov	HIGH	HIGH
Overt + insitu	HIGH	HIGH

Table 2: Experimental factors and expected judgements

the grammaticality of pro-drop and pro-drop itself having no significant effect on grammaticality in the absence of wh-movement, we expect results roughly similar to Figure 1. The interaction plot in Figure 1 represents how judgements interact with the experimental factors. The blue and red lines represent average judgements where the OVERT and NULL factors are present, while the x-axis indicates the factors corresponding to wh-movement or wh-in-situ. The plot in Figure 1 is an idealization: actual judgement data has a significant degree of variability and it is unlikely that, i.e., there would be no difference in average ratings for wh-in-situ and wh-movement with overt subjects.

Figure 1: Expected interaction plot for ratings per condition assuming the whole relevant effect being wh-movement + pro-drop.



3.4 Materials

As outlined, each experiment contained 8 diferent experimental conditions, each with a different combination of our 3 experimental factors. What follows are examples of each condition. The following examples are all from Experiment 2, e.g., second person subjects. The conditions for experiment 1 were identical except for the person feature of the subjects and the agreeing verbal

inflection.

(8) Overt subject + wh-in-situ

a. Você fez o que ontem? SINGULAR
You.S did.2S DET.M what yesterday

b. Vocês fizeram o que ontem? PLURAL
You.P did.2P DET.M what yesterday
"What did you do yesterday?"

(9) Overt subject + wh-movement

a. O que você fez ontem? SINGULAR
DET.M What you.S did.2S yesterday

b. O que vocês fizeram ontem? SINGULAR
DET.M What you.P did.2P yesterday

Both (8) and are expected to be judged as grammatical by participants. All instances of conditions with overt subjects are fully grammatical sentences in BP, as there is no reported ungrammaticality caused by wh-movement in the presence of overt subjects.

(10) Null subject + wh-in-situ

a. Fez o que ontem? SINGULAR
Did.2S DET.M what yesterday

b. Fizeram o que ontem? PLURAL
Did.2P DET.M what yesterday
"What did you do yesterday?"

(11) Null subject + wh-movement

a. *O que fez ontem? SINGULAR
DET.M what did.2S yesterday

b. Fizeram o que ontem? PLURAL
DET.M did.2P what yesterday
"What did you do yesterday?"

While (10) is expected to be judged as grammatical, (11) is reported as ungrammatical in the literature. Crucially, in (11) we have the two factor interaction we are interested in: wh-movement and pro-drop. Consequently, (11) is expected to be the lowest rating condition for both experiments.

3.5 Participants

Participants were recruited through social media posts in Portuguese on Brazilian online communities advertising the opportunity to contribute to an ongoing study about Brazilian Portuguese. Data collection occured during the final week of May 2020 for Experiment 2 and the final week of April 2021 for Experiment 1. Due to Brazilian legislation, participants could not be paid for participation. Participants were anonymous, the only identification method being a hash key associated with each individual IP address completing the survey. Before starting the experiment, participants were shown a consent letter and offered the choice to consent or not to having their judgement data collected. 269 participants were recruited for Experiment 1 and 547 participants for Experiment 2.

4 Results

The results of both experiments were analyzed in the same manner, following the methods used in Sprouse et al. (2016). Raw ratings (including fillers) for each participant were z-score transformed in order to avoid potential scale-biases. The z-score transformation has the advantage of creating a standardized scale which is comparable across participants independent of how each participant used the scale. With the z-scored judgement data, Gaussian linear mixed-effect models were built for each experiment using items and participants as random effects and WH, SUBJECT OVERTNESS and NUMBER as fixed effects. Each item and each subject identity were included in all models as intercept random effects, to account for variation in judgements among experimental items and among participants.

The models included all possible two-order interactions and a three-order interaction with all factors. Finally, the significance of each model term was assessed through sequential F-tests following standard practice as recommended in Pinheiro and Bates (2006). Through the R environment, each model was fitted using the LMER function from package LME4 (Bates et al., 2015) and sequential F-testing was done through the ANOVA function. This method allows for assessing the potential effects in altering judgement variability for each of our experimental factors.

Moreover, in order to gauge the variation in data captured by significant fixed and random effects, we calculated pseudo R^2 coefficients (Nakagawa and Schielzeth, 2013) for the most significant model[4] in each experiment. Pseudo R^2s can be partitioned into Marginal and Conditional values. The marginal value represents the variation explained by fixed effects (e.g., the experimental factors), while the conditional value represents the variation explained by fixed and random effects (e.g., the entire model including random effects).

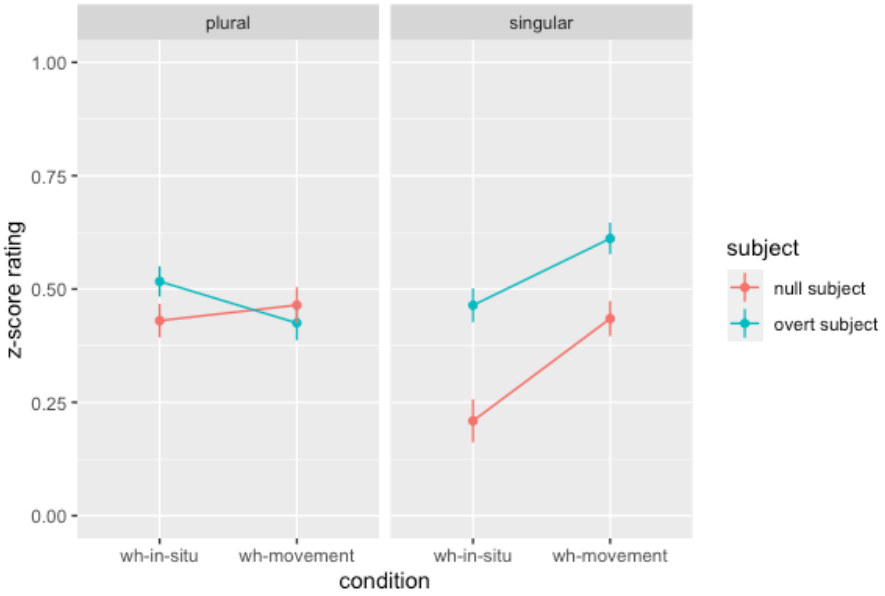
4.1 Experiment 1: first person pro-drop

For 1st person singular subjects, participants showed a preference for overt subjects and for questions with wh-movement, the OVERT + WHMOV condition being the highest rated of all four, and NULL + WHSITU the lowest, Figure 2. Furthermore, model testing revealed a near-significant effect ($p = 0.0487$) in acceptability for subject overtness. The data on 1st person plural subjects showed no significant effects apart from random effects. The results from sequential F testing of each factor and factor combination can be found in Table[5] 3.

[4]For that purpose, we selected the model with only significant fixed effects as informed by sequential F-testing of the model with all effects

[5]Not shown in Table 1 is the DenDF for all factors, which was of 54.5412

Figure 2: Interaction plot for ratings per condition, 1st person subjects



Importantly, judgements remained largely on the grammatical side of the scale (e.g., above the 0 line), as it can be observed in the rating density per condition (Figure 3)[6] and the box-plot of individual ratings across conditions (Figure 4).

	Sum Sq	Mean Sq	F value	Pr(>F)
number	0.0443	0.0443	0.1315	0.7182
overtness	1.3669	1.3669	4.0549	0.0487
wh	0.5625	0.5625	1.6687	0.2016
overtness:wh	0.0791	0.0791	0.2345	0.6300
number:overtness	0.5093	0.5093	1.5110	0.2240
number:wh	1.0203	1.0203	3.0268	0.0872
number:overtness:wh	0.0112	0.0112	0.0331	0.8563

Table 3: Results from sequential F-testing for the model fitted using the data from Experiment 1

[6]The codes for each condition have a letter indicating singular or plural P/S followed by nons (overt subject) or ns (pro-drop) and whsitu (wh-in-situ) or whmov (wh-movement).

Figure 3: Rating density per condition, 1st person subjects

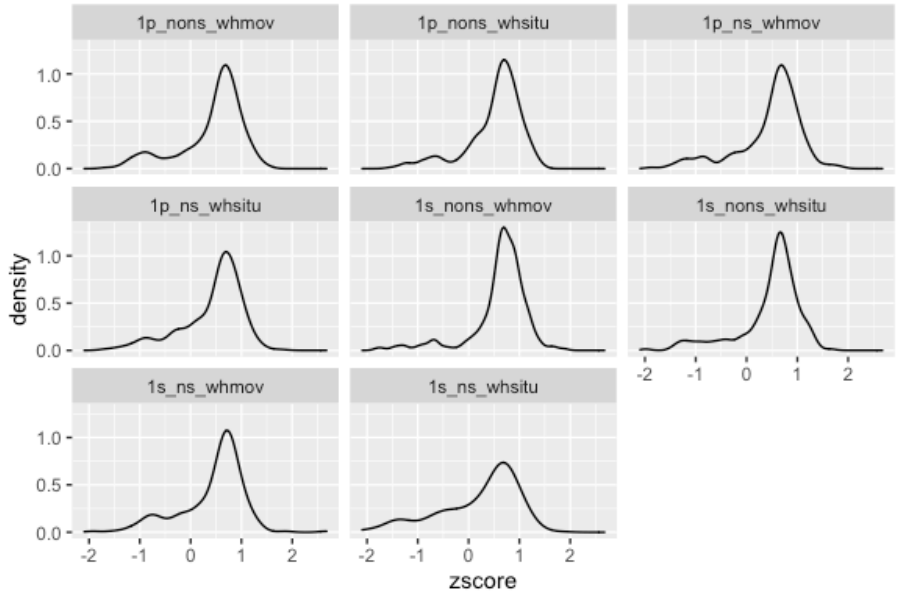
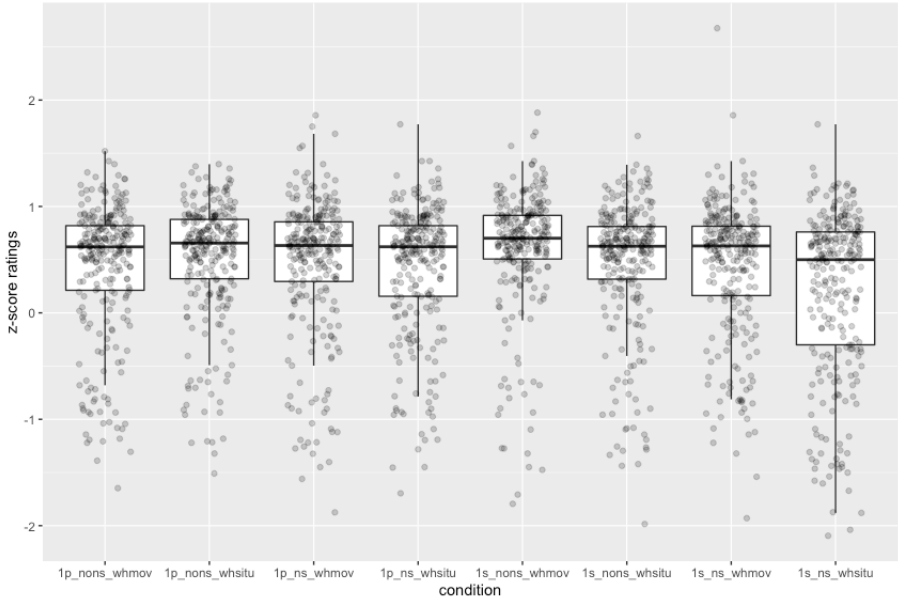


Figure 4: Ratings across conditions, each dot represents an individual rating, 1st person subjects



Finally, to gauge how well the model represented variability, pseudo R^2 were calculated for a model with just random effects and WH and yielded a conditional value of 0.167 and a marginal value of 0.007. The conditional value indicates that the model with all factors explained a pro-

portion of 0.167 of the variation in the data. The marginal value indicates that the fixed factors alone explained a proportion of 0.007 of the variation in the data.

4.2 Experiment 2: second person pro-drop

For 2nd person singular subjects, participants showed a preference for overt subjects and a dis-preferrence for pro-drop with wh-movement. The lowest rating condition was the NULL + WHMOV condition.

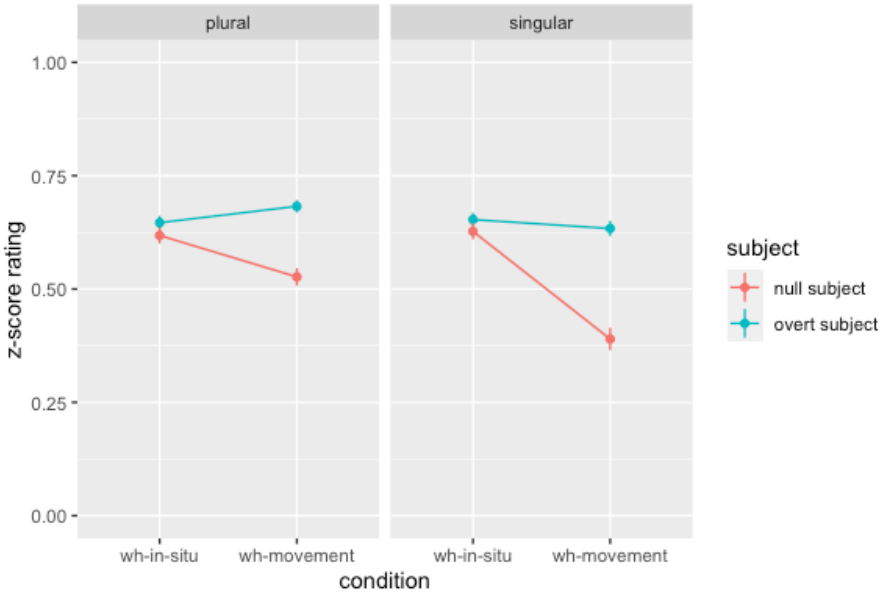
Moreover, the experiment revealed a statistically significant effect for subject overtness (p = 0.0006), wh-movement (p = 0.0176) and the two-order interaction between wh-movement and subject overtness (p = 0.0056). The results from sequential F testing of each factor and factor combination can be found in Table[7] 4

	Sum Sq	Mean Sq	F value	Pr(>F)
number	0.3067	0.3067	2.0170	0.1612
overtness	2.0090	2.0090	13.2115	0.0006
wh	0.9111	0.9111	5.9911	0.0176
subject:wh	1.2623	1.2623	8.3007	0.0056
number:overtness	0.1257	0.1257	0.8265	0.3673
number:wh	0.2219	0.2219	1.4592	0.2322

Table 4: Results from sequential F-testing for the model fitted using the data from Experiment 2

[7]Not shown in Table 1 is the DenDF for all factors, which was of 54.9121

Figure 5: Interaction plot for ratings per condition, 2nd person subjects



Again, judgements remained largely on the grammatical side of the scale (Figures 6 and 7).

Figure 6: Rating density per condition, 2nd person subjects

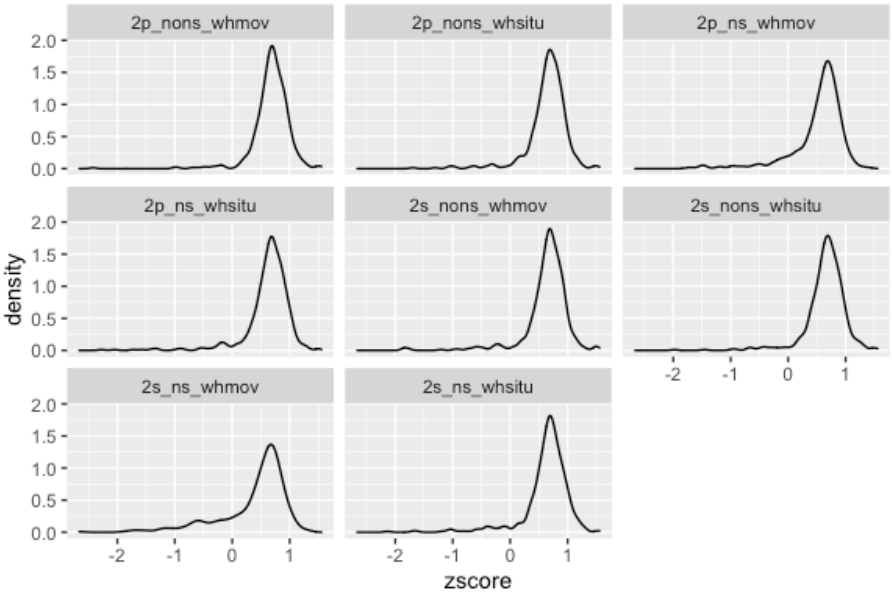
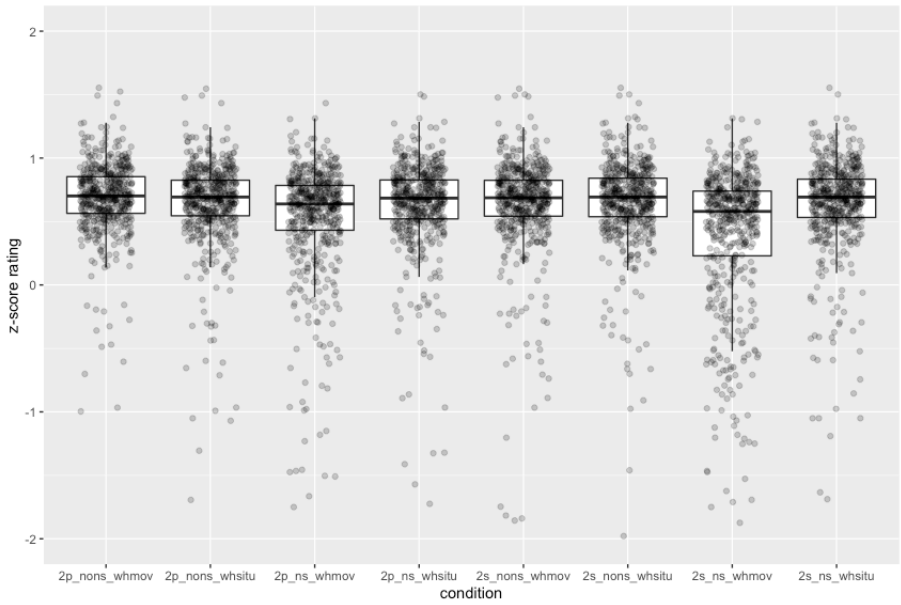


Figure 7: Ratings across conditions, each dot represents an individual rating, 2nd person subjects



As in Experiment 1, R^2 values were calculated for the model with the factors WH, OVERTNESS and the interaction between both. The marginal value found was of 0.0038 and the conditional value found was of 0.1539.

5 Discussion

The first crucial observation concerning the experimental data is that there is no significant evidence for any factor or combination of factors contributing for ungrammaticality per-se. As mentioned, in both experiments all conditions were rated largely around the top-end of the scale and no condition displayed ratings centered around the lower-end of the scale, as would be expected of an ungrammatical structure. This finding contradicts the literature. As outlined in section 2, the literature on BP reports matrix pro-drop as ungrammatical whenever wh-movement is present, e.g., the NULL + WHMOV condition. There isn't, however, statistical evidence in our data that speakers considered this structure ungrammatical.

In that context, our first conclusion is that 1st and 2nd person matrix pro-drop in BP seems

to be less a matter of grammaticality per-se but of speakers' preference. Moreover, the R^2 values found for both experiments also reveal that a large proportion of the data variation was not explained by random or fixed factors in our models. This fact points to judgement variability arising from potentially extra-grammatical factors or from interactions between random effects not accounted for.

	Marginal R^2	Conditional R^2
Experiment 1	0.003	0.153
Experiment 2	0.007	0.167

Table 5: R^2 values for both experiments

These facts are all congruent with most findings in the diachronic and sociolinguistic literature. Pro-drop is still part of an ongoing change in the language and while there is a preference for overt subjects, it is still a preference for an equally grammatical option. All conclusions arising from the data in this study are about preferences within grammatical structures and not about the grammaticality versus ungrammaticality of the given structures. Moreover, there seems to be a great degree of variability related to factors beyond syntactic structure, e.g., sociolinguistic and discoursive factors which have not been accounted for in the experimental design. In the following sections I discuss some of the main possible conclusions that can be drawn from the experimental data.

5.1 2nd person subjects

Going back to our hypotheses, repeated below, we come to mixed conclusions. Hypothesis A was confirmed for second person subjects: the lowest rated condition was the pro-drop + wh-movement condition and we found a statistically significant effect on acceptability for the interaction between subject overtness and wh-movement. Moreover, we found no statistically significant effect for number, pointing to Hypothesis B being correct. Generally, Experiment 2 reflected

largely the expected data pattern, even though the judgement data itself did not, e.g., the ratings remained in the grammatical side of the scale.

(12) **Hypothesis A: pro-drop and wh-movement**

We expect the lowest judgements for conditions with pro-drop + wh-movement.

(13) **Hypothesis B: number and pro-drop**

We expect number to play no role in increasing or reducing judgements for all conditions.

5.2 1st person subjects

The data for 1st person subjects did not provide much information regarding acceptability. We found no statistically significant effects apart from a near-significant effect for subject overtness. Surprisingly, for singular subjects, participants showed a preference for wh-movement, and the condition we expected to be rated the lowest - NULL + WHMOV - was rated the highest, against Hypothesis A. For plural subjects no conclusions can be drawn. The data shows only random effects playing any role in judgements, and R^2 values indicate a very small (0.007) proportion of variability being explained by our fixed effects. In a broad sense, it seems there is a great degree of variability unaccounted for and no clear observable effects related to our experimental factors. In line with Hypothesis B, number seemed statistically insignificant.

6 Final remarks

We found no evidence supporting the grammaticality judgement data reported in the literature for pro-drop in matrix questions in BP.. Namely, we have not find experimental evidence for 1st and 2nd person pro-drop being ungrammatical with wh-movement. We found evidence speakers *prefer* overt subjects in the presence of wh-movement for 2nd person subjects, but our judgement

data does not point to this preference being caused by ungrammaticality.

Furthermore, we have found no evidence that number plays any role in the acceptability of pro-drop. In both experiments we found no significant effect associated with number.

Moreover, we found evidence for the novel observation that person seems to play an important role in the acceptability of pro-drop in BP: the results from both experiments differed significantly. Specifically, while 2nd person pro-drop showed results in line with our working hypotheses, 1st person subjects showed unexpected patterns for singular subjects. For 1st person pro-drop, our data points to speakers preferring pro-drop with wh-movement, a structure which is reported as ungrammatical in current literature.

In a general sense, this study provided evidence for a more flexible picture of pro-drop in matrix questions in BP. Speakers show preferences for certain structures, but said preferences do not reflect grammaticality. In that regard, the results from this experiment require a revision of the current empirical landscape and, consequently, some reformulations of the current approaches to pro-drop in BP.

On one hand, the results bring evidence that there is no reason to believe that wh-movement interacts with the grammaticality of pro-drop in BP. The (reported) interaction between pro-drop and wh-movement caused problems for current approaches to pro-drop in partial NSLs. These problems cease to exist if we conclude that the contrast between wh-movement and wh-in-situ regarding pro-drop reflects a preference but not a difference in grammaticality.

On the other hand, the conclusions reached in this study don't align necessarily with any current account of pro-drop in BP. One of the core arguments of the topic drop analysis described in section 2.4 is that wh-movement creates a minimality violation which blocks movement of the subject to its higher topic position. If pro-drop is indeed grammatical with wh-movement this argument is no longer empirically supported.

Finally, if our results point to effects which reflect preferences for certain syntactic structures we must ask which kind of theory explains these preferences. An easy solution is postulating that

the effects found in this study reflect facts of *performance* in the sense of mainstream generativism (Chomsky, 2014) and are beyond the scope of syntactic theory. While this answers part of the question it does not explain which factors condition the effects observed in this study. Ideally, the answer to this question would provide a model to account for how speakers select preferred syntactic structures. If, as we found in Experiment 2, speakers prefer pro-drop with wh-in-situ in matrix questions we may ask if this preference is reflected in analogous structures (i.e., embedded clauses). Moreover, the potential influence of sociolinguistic factors could be directly related to the patterns observed. The generally low R^2 values found for both experiments point to a reality where our experimental factors played little role in explaining the data variability. This suggests the potential influence of extra-syntactic factors in speakers' judgements. While this study did not collect any kind of sociolinguistic information (e.g., region the participants lived, age, gender, social extract, education level), it seems fruitful for future work to collect this kind of information to asses if and how much sociolinguistic factors play a role in speakers' preferences. Apart from sociolinguistic factors, it is likely discoursive information plays a role in speakers' preferences. The experimental stimuli did not include discourse fragments to contextualize each target sentence, making it so that all potential discoursive factors were left out of our modeling. For future work, discoursive information could be taken into account when designing similar experiments.

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